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WAH-*i*: Optimising Microphone Array Geometry for Customised Localisation Accuracy

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Abstract.

1. Accurate spatial localisation of free-flying echolocating bats is foundational for resolving fine-scale flight behaviour, prey interception, and spatial decision-making in natural environments. Acoustic localisation using microphone arrays is widely employed for this purpose, yet array geometries in field studies are typically chosen heuristically rather than systematically optimised. As portable multichannel ultrasonic recording systems become increasingly accessible, principled design guidelines are needed to ensure reliable localisation performance under practical deployment requirements.

2. I introduce an iterative array optimisation algorithm that designs microphone geometries by maximising localisation reliability within a predefined three-dimensional field of interest. The method evaluates candidate geometries using simulated acoustic emissions and time-difference-of-arrival localisation, quantifying performance as a volumetric pass rate: the proportion of source locations that meet a user-defined accuracy threshold. Microphone positions are iteratively perturbed and accepted based on improvements to this task-level metric, while enforcing practical constraints on array aperture, inter-sensor spacing, and deployability.

3. Across canonical polyhedral geometries, random initialisations, and arrays comprising four to twelve microphones, optimisation consistently produced rapid early gains followed by convergence to geometry-specific performance limits. Under fixed-aperture constraints, increasing the microphone count yielded diminishing returns, and optimised low-order arrays — particularly four-microphone configurations — matched or exceeded the volumetric localisation performance of higher-order arrays with suboptimal geometry. Analysis of optimisation trajectories further revealed that convergence dynamics scale with array order, whereas achievable volumetric performance is dominated by geometry rather than sensor number.

4. These results demonstrate that array geometry is the primary determinant of volumetric localisation reliability, and that efficient, portable arrays can be systematically designed using optimisation rather than heuristic rules. The proposed framework is broadly applicable to bioacoustic localisation problems beyond echolocating bats, including avian tracking, passive acoustic monitoring, and conservation-oriented sensing, and provides a general approach for designing task-optimised acoustic sensor arrays for a wide range of applications.

Keywords: acoustic localisation, array geometry, bat echolocation, microphone arrays, optimisation, TDOA

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1 INTRODUCTION

Echolocating bats navigate, forage, and interact in complex three-dimensional environments by emitting ultrasonic calls and analysing the returning echoes [1–4]. This active sensing modality enables precise spatial perception without relying on vision. As a result, bats have become a central model system for studying acoustically-guided spatial behaviour, sensory ecology, and predator–prey interactions in the wild. Quantitative investigation of these behaviours, however, critically depends on the ability to localise animals accurately in three dimensions over ecologically relevant spatial scales.

Many behavioural questions in bat ecology require centimetre-scale localisation accuracy across observation volumes spanning several cubic metres. Examples include reconstructing flight trajectories during prey capture, estimating spatial decision rules during obstacle avoidance, and resolving fine-scale interactions between conspecifics or between bats and prey. Achieving such precision in field conditions is challenging. Optical tracking systems are limited by lighting requirements, line-of-sight constraints, and restricted working volumes, while satellite-based positioning lacks the spatial and temporal resolution required for small, fast-moving animals. Acoustic localisation using microphone arrays, therefore, represents the most practical and widely adopted approach for tracking echolocating bats in natural environments.

Microphone-array systems exploit time-differences-of-arrival (TDOA) of ultrasonic calls across spatially distributed sensors to estimate source position. This approach is well-suited to nocturnal field studies, provides high temporal resolution when combined with high sampling rates, and can be tailored to species-specific call characteristics. A range of array-based tracking systems has been used, including indoor and outdoor installations in experiments that have addressed a variety of ethological questions in bats, such as [5–16]. Further, microphone arrays find broader applications of acoustic localisation in behavioural ecology, monitoring and conservation [17–24]. Despite these advances in applications, the geometric design of microphone arrays in most biological applications remains largely *ad hoc*, guided by heuristic symmetry or practical convenience rather than systematic optimisation.

Field deployments impose additional constraints that complicate array design. The number of microphones is limited by cost, synchronisation complexity, power consumption, and data throughput, often restricting systems to fewer sensors. Physical constraints, such as mounting structures, portability, and microphone placement, further limit design choices. Environmental factors, including wind, temperature gradients, and multipath reflections, introduce additional sources of localisation error. These constraints create a fundamental trade-off: increasing array aperture improves geometric precision, but larger arrays are harder to deploy and maintain. This motivates a central practical question: given a fixed microphone budget and physical constraints, what array geometry maximises localisation reliability over a target observation volume?

From a mathematical perspective, TDOA-based localisation is inherently geometry-dependent. Relative arrival times are estimated from cross-correlations of microphone signals and converted into source position estimates via multilateration [25–29]. While timing precision depends on factors such as sampling rate and signal-to-noise ratio [25, 27, 30], the resulting position error is strongly modulated by sensor geometry [26, 31]. For a fixed timing uncertainty, the same array can yield highly accurate localisation in some regions of space and catastrophic errors in others. This spatial heterogeneity is commonly described by geometric dilution of precision (GDOP),

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which captures how geometry amplifies timing errors as a function of source position.

A substantial body of work has addressed array optimisation using analytical performance measures. Approaches based on minimising the Cramér–Rao lower bound (CRLB) maximise Fisher information associated with sensor placement [32–34], while related methods minimise the GDOP at selected target locations [31, 35]. Although theoretically elegant, these approaches optimise precision at specific points or weighted sets of points, rely on assumptions such as Gaussian noise and local linearity, and provide limited insight into volumetric reliability under hard success–failure criteria. In practice, biological localisation systems operate with threshold-based performance requirements, strong near-field effects, and non-Gaussian error distributions that are poorly captured by purely analytical metrics.

More general optimisation frameworks, including convex and semidefinite programming methods [36, 37], offer elegant solutions under relaxed assumptions but struggle to accommodate non-convex constraints such as minimum inter-sensor spacing, bounded array extent, and the requirement for non-coplanar three-dimensional layouts. Coverage-based sensor placement methods developed in robotics and surveillance [38, 39] typically focus on two-dimensional detection or visibility problems and do not directly address the timing precision and volumetric error structure inherent to acoustic multilateration.

Consequently, several gaps remain. First, most optimisation studies focus on pointwise or averaged error metrics rather than task-level measures of volumetric reliability. Second, little attention has been paid to geometry-dependent performance ceilings: whether different array topologies impose fundamental limits that cannot be overcome by coordinate refinement. Third, optimisation outcomes are rarely assessed statistically across repeated runs, leaving it unclear whether observed solutions are robust or artefacts of specific initialisations. Finally, validation of optimisation strategies using signal parameters representative of bat echolocation needs to be evaluated.

Here, I address these gaps by framing microphone-array design as a constrained, task-oriented optimisation problem. Rather than minimising average localisation error, I evaluate performance using a *volumetric pass rate*: the fraction of source locations within a three-dimensional observation volume that can be localised to within a predefined accuracy threshold. This formulation transforms array design into a nonlinear optimisation over microphone coordinates within a compact feasible set defined by physical constraints on array extent and inter-sensor spacing. While the existence of at least one optimal configuration is guaranteed by standard results from analysis (the Weierstrass extreme value theorem [40, 41]), this guarantee is purely existential. The resulting optimisation landscape is highly non-convex due to GDOP singularities, near-field ambiguities, and discrete spatial constraints, precluding closed-form solutions and motivating direct, derivative-free search strategies.

To explore this landscape, I extend the Array Widefield Acoustics Heuristic (Array WAH) [42, 43] to an inverse, optimisation-driven framework termed the Widefield Acoustics Heuristic - *inverse iterative* (WAH-i). WAH-i employs a deterministic coordinate-search strategy that iteratively perturbs microphone positions to maximise volumetric pass rate under realistic field and geometric constraints, using the same forward localisation and evaluation machinery as Array WAH. Using signal parameters representative of aerial-hawking bats, I evaluated six canonical polyhedral geometries ranging from tetrahedral to icosahedral layouts, alongside randomly initialised arrays. Optimisation was treated as a stochastic process rather than a single-shot design

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exercise. For each configuration, five independent optimisation runs were performed, yielding 55 optimisation trajectories in total. This batch-based approach enables separation of optimisation dynamics from geometry-imposed performance limits and allows statistical assessment of convergence behaviour, robustness, and geometry drift.

The results demonstrate that microphone array performance is governed less by sensor count *per se* than by how effectively a given geometry exploits the constraints of the task and sensing volume. By framing array optimisation in terms of volumetric reliability, economy, and responsiveness, this study provides a principled alternative to heuristic or scale-driven design choices that have traditionally guided bioacoustic localisation systems. The finding that carefully optimised low-order arrays can match or exceed the usable performance of denser configurations has immediate practical implications for field deployments, where portability, mechanical stability, power consumption, and ease of replication are often limiting factors.

In a broader context, the task-oriented framework introduced here is readily transferable to other active and passive acoustic sensing applications, including robotics, human–machine interaction, and distributed sensor networks, where similar trade-offs among accuracy, robustness, and system complexity arise. Future developments may extend this approach to dynamic or adaptive array configurations, motion-aware optimisation objectives, and learning-based surrogates that exploit the optimisation data generated here to accelerate design exploration. Together, these directions point toward advancing acoustical sensing systems in which array geometry is not fixed by convention but is explicitly shaped by the informational demands of the task at hand.

2 METHODS

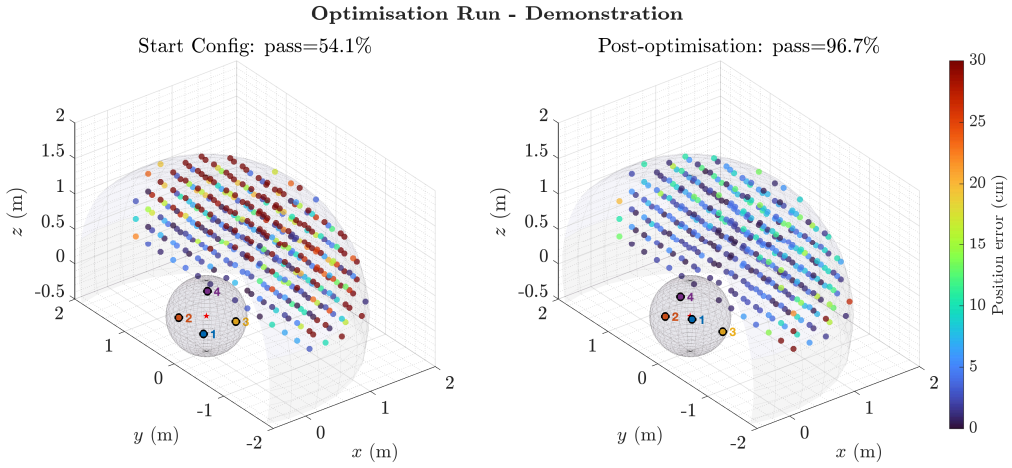
This study combines simulation-based localisation analysis with an iterative, performance-driven optimisation framework to compute an optimised geometric configuration of a microphone array for a desired field of spatial localisation accuracy. All methods are formulated at the level of task-relevant performance within a defined three-dimensional field of accuracy. Using synthetic echolocation signals and a localisation pipeline, array geometries are systematically evaluated, perturbed, and re-evaluated to characterise both achievable accuracy limits and optimisation dynamics under realistic spatial and mechanical constraints. The following sections detail the signal model, localisation procedure, field-of-accuracy definition, optimisation algorithm, and analysis metrics used throughout the study.

2.1 Optimisation formulation: task-level array design for volumetric TDOA localisation

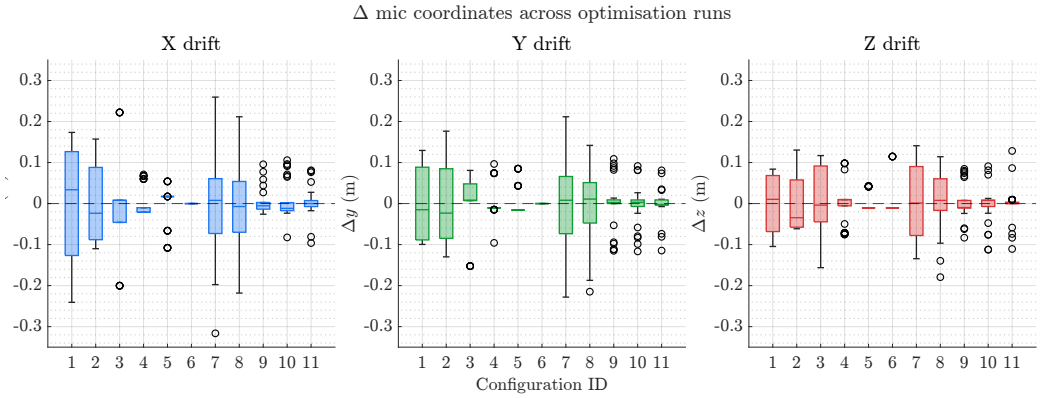
I considered passive three-dimensional acoustic localisation of echolocation calls using time-difference-of-arrival (TDOA) measurements from a compact microphone array. Let an array configuration be represented by a matrix $\mathbf{M} \in \mathbb{R}^{N\times 3}$, whose rows $\mathbf{m}_i = [x_i, y_i, z_i]^\mathsf{T}$ give the Cartesian coordinates of the i-th microphone. Array performance was evaluated over a discrete set of candidate source positions $\mathcal{G} \subset \mathbb{R}^3$ sampling the target observation volume. For a source at $\mathbf{s} \in \mathcal{G}$, localisation yields an estimate $\widehat{\mathbf{s}}$, and I define the Euclidean position error

$$e(\mathbf{s}) \;=\; 100\,\|\widehat{\mathbf{s}} - \mathbf{s}\|_2 \quad [\mathrm{cm}]. \tag{1}$$

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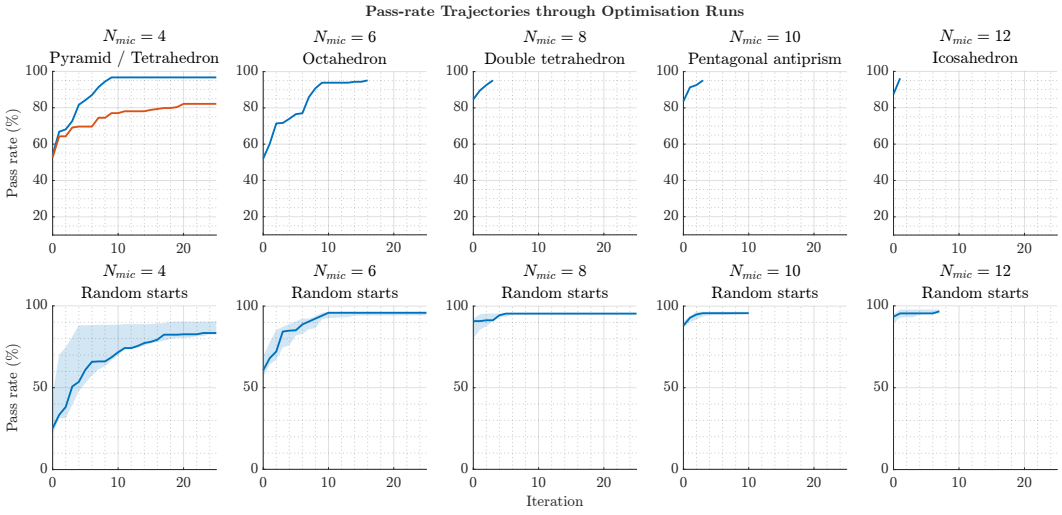


(a) Example localisation error map before and after optimisation for one selected geometry, pyramid.



STD: 1=Pyramid (N_{mic}=4); 2=Tetrahedron (4); 3=Octahedron (6); 4=Double tetrahedron (8); 5=Pentagonal antiprism (10); 6=Icosahedron (12)
RND: 7=Random (4); 8=Random (6); 9=Random (8); 10=Random (10); 11=Random (12)

(b) Distribution of per-microphone coordinate changes ($\Delta x, \Delta y, \Delta z$) between the initial and final configurations, summarised by configuration ID.



(c) Pass-rate (%) across iterations, shown separately for standard initial geometries (top row) and random starts (bottom row) for each microphone count N_{mic} group.

Figure 1: Optimisation framework and outcome. Microphone-array geometry was optimised by iteratively updating microphone coordinates to maximise a volumetric *pass-rate* (defined as the fraction of test locations meeting the predefined localisation-accuracy criterion of 15 cm), starting either from standard geometric templates or from random initialisations. (a) A representative run illustrates how the spatial distribution of localisation error changes from the initial pyramid configuration to the post-optimisation configuration. (b) Across batches of runs, optimised solutions exhibit systematic coordinate shifts, summarised as per-axis microphone drift relative to the starting geometry. (c) The optimisation yields rapid early gains in pass rate followed by convergence, with trajectories shown for each N_{mic} under standard and random starting conditions. (*Orange - tetrahedron*)

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Because field tracking often hinges on whether localisation meets an operational accuracy requirement rather than on mean error alone, I used a task-level *pass–fail* criterion: a source location was deemed a *pass* if

$$e(\mathbf{s}) \leq \theta, \tag{2}$$

with acceptance threshold $\theta = 15$ cm. The primary performance metric was the *volumetric pass rate*, defined as the fraction of valid evaluation points meeting the criterion:

$$\mathrm{PassRate}(\mathbf{M}) \;=\; \frac{\#\{\mathbf{s} \in \mathcal{G}_{\mathrm{valid}} : e(\mathbf{s}) \leq \theta\}}{\#\mathcal{G}_{\mathrm{valid}}}. \tag{3}$$

Here $\mathcal{G}_{\mathrm{valid}}$ denotes the subset of evaluation points retained after considering the exclusion conditions (e.g. source positions too close to microphones; see below). In addition to positional error, I computed an angular error relative to a reference microphone by comparing estimated and true azimuth–elevation directions and reporting the Euclidean difference in the $(\mathrm{az}, \mathrm{el})$ plane (degrees). Position error was used for optimisation and for the pass criterion, while angular error served as a complementary performance descriptor (analysed in Fig. 3 and Fig. 5).

Array design was posed as a constrained nonlinear optimisation problem: find microphone coordinates $\mathbf{M}$ that maximise localisation reliability over $\mathcal{G}$ subject to physical boundary constraints. Specifically, all candidate geometries were required to satisfy (i) a bounded aperture constraint,

$$\|\mathbf{m}_i\| \leq L_{\max} \quad \forall i, \tag{4}$$

with $L_{\max} = 0.50$ m; and (ii) a minimum inter-microphone spacing constraint,

$$\|\mathbf{m}_i - \mathbf{m}_j\| \geq d_{\min} \quad \forall i \neq j, \tag{5}$$

with $d_{\min} = 6.25$ cm. Candidate geometries were also required to be non-coplanar (numerically rank-3 after centring), ensuring a fully three-dimensional array capable of resolving elevation.

Although these constraints define a bounded set of admissible array geometries, the resulting optimisation problem is strongly non-convex. Small changes in microphone placement can produce abrupt changes in localisation performance due to geometric dilution of precision, near-field effects, and occasional failures in time-difference-of-arrival estimation and nonlinear multilateration. In addition, performance was evaluated using a discrete pass–fail criterion rather than a smooth cost function, further fragmenting the objective landscape. As a result, the optimisation surface contains multiple local optima and discontinuities, making gradient-based or convex methods unsuitable and motivating the use of direct, derivative-free search strategies.

2.2 Evaluation grid and spatial sampling of the observation volume

Performance was evaluated on a three-dimensional lattice of candidate source positions with step size 0.25 m, spanning a shell-like region around the array centre. The outer radius was $R = 2.0$ m and an inner exclusion radius $R_{\mathrm{in}} = 0.30$ m prevented evaluation arbitrarily close to the array. The inner exclusion is important because both acoustic time-delay estimation and multilateration can become ill-conditioned in extreme near-field scenarios (e.g. when the source is close to one sensor). In addition, I restricted evaluation to a forward, upper sector of space to reflect typical field arrangements in which the array is placed below or beside the flight volume;

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thus, not all spatial octants contributed equally to the objective.

For each evaluation point, localisation was performed only if the source-to-nearest-microphone distance exceeded the minimum safety distance (0.08 m). Points failing this criterion were excluded from summary statistics and pass-rate calculation. Numerical failures (e.g. invalid delays or solver non-convergence) at otherwise valid points were treated as catastrophic errors by assigning a large fixed penalty error; this prevents such cases from being silently ignored while ensuring that failures reduce the pass rate.

2.3 Forward acoustics, TDOA estimation, and multilateration

To test array performance under realistic signal conditions, I simulated frequency-modulated echolocation calls with parameters representative of insectivorous bats: a 70–30 kHz FM downsweep, sampling rate $f_s = 192$ kHz, duration 5 ms, and additive noise producing an effective signal-to-noise ratio of 40 dB. For a source at $\mathbf{s}$, each microphone signal was generated by propagating the call with a fixed speed of sound $c = 343\ \mathrm{m\,s^{-1}}$ and applying the corresponding time-of-flight delay. TDOAs were estimated relative to a designated reference microphone using normalised cross-correlation, with the delay chosen as the lag that maximised the correlation magnitude. These estimated delays define a set of range differences of the form

$$\|\mathbf{m}_i - \mathbf{x}\| - \|\mathbf{m}_1 - \mathbf{x}\| \approx c\,\widehat{\tau}_i, \quad i = 2, \ldots, N, \tag{6}$$

where $\mathbf{x}$ is the unknown source position and $\widehat{\tau}_i$ is the estimated TDOA relative to microphone 1. The source estimate $\widehat{\mathbf{s}}$ was obtained by solving the resulting nonlinear least-squares problem (hyperbolic multilateration). This procedure yields a localisation error vector over $\mathcal{G}$, from which pass rate and distributional summaries are derived. The implementation was built on the Widefield Acoustics Heuristic toolkit [43] (See Umadi [42] for the description of the original methodology).

2.4 Objective function for geometry optimisation

Optimisation targeted volumetric reliability while discouraging solutions that achieve a pass rate by tolerating large errors at the remaining points. The objective function was formulated as a goal-oriented scalarisation of competing criteria. Following the standard approaches in multi-objective optimisation [44, 45], pass rate and mean localisation error were combined into a single objective using a linear weighting. To prevent domination by extreme outliers, the mean error term was normalised by a characteristic failure scale and capped, consistent with robust loss formulations used in inverse problems [46, 47]. This construction reflects the operational goal of maximising volumetric reliability while weakly discouraging solutions that achieve high pass rates by tolerating catastrophic errors at a minority of points. I therefore maximised a scalar objective combining pass rate with a weak penalty on mean positional error,

$$J(\mathbf{M}) \;=\; \mathrm{PassRate}(\mathbf{M}) \;-\; \lambda\, \min\!\left(\frac{\overline{e}(\mathbf{M})}{e_{\mathrm{fail}}},\, 1\right), \tag{7}$$

where $\overline{e}(\mathbf{M})$ is the mean error over valid points, e_{fail} is the fixed penalty assigned to catastrophic failures, and λ is a small weighting factor. This form prioritises increasing the fraction of the working volume meeting the operational threshold while retaining sensitivity to broad error

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inflation and failure modes.

The choice of λ governs the trade-off between *coverage* and *accuracy margin*. For small values of λ, optimisation is dominated by pass-rate maximisation, favouring geometries that bring as many spatial locations as possible below the acceptance threshold, even if the resulting error distribution remains sharply peaked near that threshold. Increasing λ places greater emphasis on reducing the mean error among passing locations, encouraging solutions with improved robustness and larger safety margins at the expense of marginal reductions in pass rate. Importantly, because the penalty term is normalised and clipped, λ primarily influences behaviour near the pass–fail boundary rather than the treatment of extreme outliers.

The effectiveness of this trade-off is coupled to the spatial resolution of the coordinate-search procedure. Microphone positions are updated using a bounded coordinate descent with a linearly decaying step size,

$$\Delta(t) \;=\; L_{\max}\,\max\!\left(\text{minStepFrac},\; \text{initStepFrac}\left(1-\frac{t-1}{T-1}\right)\right), \tag{8}$$

where T is the total number of optimisation iterations. Early iterations permit large exploratory movements that allow the optimiser to escape poor initial configurations, whereas later iterations perform fine-scale adjustments to refine geometry within a locally optimal basin.

Larger values of λ therefore benefit from sufficiently small terminal step sizes, as improvements in mean error typically require precise, local rearrangements of microphone positions rather than coarse geometric changes. Conversely, overly small initial step sizes can prevent the optimiser from escaping suboptimal basins regardless of penalty weighting. Together, λ and the step-size schedule define a coupled control mechanism that determines whether optimisation prioritises volumetric coverage, accuracy margin, or a balanced compromise between the two (see Appendix B)

2.5 The coordinate-search using the *inverse iterative* of Widefield Acoustics Heuristic (WAH-*i*)

I optimised the array geometry using a deterministic direct search strategy. The WAH-i algorithm performs iterative coordinate refinement by proposing candidate displacements for individual microphones, evaluating each candidate on the full grid $\mathcal{G}$, and accepting the candidate that yields the greatest improvement in the objective J. This procedure is repeated across microphones in a cyclic schedule, ensuring that all sensors are revisited as the objective landscape evolves. An arbitrary pass-rate ceiling was set at 95%.

Candidate moves are generated along a fixed set of probe directions that include Cartesian axes and diagonals, as well as radial inward/outward directions relative to the array centre. Step size is governed by a scheduled decay: early iterations explore the space using larger steps to quickly exploit obvious improvements, whereas later iterations use smaller steps to refine local structure (see above). Candidate geometries are re-centred after each move to maintain a consistent array centre, and any microphone exceeding the aperture bound is projected back onto (or just within) the allowed maximum radius. Candidate moves violating the minimum spacing or non-coplanarity constraints are rejected without evaluation.

To avoid stagnation in shallow local optima, WAH-i includes a stall-detection mechanism. If successive iterations fail to improve the objective beyond a small tolerance, a stochastic per-

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turbation (*shake*) is applied to the full geometry to promote escape from local basins. The perturbed geometry is subsequently repaired by projection and boundary checks; if the physical conditions cannot be restored, a new non-coplanar geometry is drawn. Optimisation terminates when a target pass rate is reached or when the maximum iteration budget is exhausted.

2.6 Experimental conditions and repeated-run design

I evaluated the optimisation behaviour across a set of canonical three-dimensional array geometries spanning microphone counts $N_{mic} = 4$ to $N_{mic} = 12$, alongside randomly initialised arrays at matched N_{mic}. Canonical configurations comprised polyhedral layouts commonly used as "reasonable" geometric baselines in array design (e.g. tetrahedral, pyramidal, octahedral, antiprismatic, and icosahedral topologies). Random starts provide a practical reference for performance achievable without topology-informed initialisation.

Because direct-search optimisation in a non-convex landscape can be sensitive to initialisation and stochastic escape events, each configuration was optimised using five independent runs. This repeated-run design enables separation of geometry-imposed performance ceilings from run-to-run optimisation variability, and supports statistical evaluation of convergence behaviour (Fig. 1c and Fig. 2).

2.7 Analysis of optimisation dynamics and localisation performance

Optimisation outcomes were analysed at three complementary levels, each corresponding to a figure set reported in the Results.

Trajectory-level convergence and saturation. For each configuration, I extracted pass-rate trajectories across iterations and summarised them across repeats using robust central tendency and uncertainty bands. To ensure consistent alignment, intermediate perturbation steps introduced during stall escapes were excluded from trajectory summaries. These analyses quantify rapid early gains and geometry-dependent saturation behaviour (Fig. 1c).

Runtime diagnostics and geometry drift. To determine whether pass-rate differences reflect optimisation inefficiency or intrinsic geometric limits, I compared (i) the maximum pass rate achieved during a run, (ii) the final pass rate, and (iii) the iteration at which 95% of the run-wise maximum was first reached (Fig. 2). Finally, to quantify how strongly optimisation altered microphone coordinates, I computed per-microphone displacement vectors between pre- and post-optimisation states and summarised their component-wise distributions across configurations (Fig. 1b). These diagnostics allow assessment of whether high performance requires large geometric rearrangements or arises from modest topology-preserving adjustments.

Spatial structure: error versus distance. To quantify how performance varies with working range, I binned grid points by radial distance from the array centre and computed median error and a 10–90% band as a function of distance. This analysis captures geometry-dependent differences in the onset of failure at larger radii and the persistence of distance-dependent degradation under a fixed aperture constraint (Fig. 3).

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Distributional performance before vs. after optimisation. For each run, localisation errors were evaluated on the same grid at the initial state and at the selected post-optimisation state. I summarised the redistribution of error using average-run histograms (Fig. 4) and robust statistics including the 95th percentile (p_{95}) to characterise tail behaviour and catastrophic failures (Fig. 5). All error summaries were computed over valid grid points only.

2.8 Software implementation and user-accessible optimisation tool

To facilitate reproducible exploration of microphone array configurations and to lower the barrier to applying the proposed optimisation framework, the algorithm was packaged as a user-accessible software tool. The implementation allows users to explore array geometries with user-defined accuracy criteria and spatial performance requirements, enabling rapid assessment of how geometric design choices influence localisation reliability across a three-dimensional observation volume.

The core optimisation and simulation pipeline was implemented in MATLAB R2023b (MathWorks, Natick, MA, USA). In addition to full programmatic access via documented functions and scripts, the framework is distributed as an interactive graphical application that exposes key parameters, including microphone count, initial geometry, aperture constraints, optimisation settings, and accuracy thresholds. This design allows both scripted batch analyses and exploratory, user-driven investigation of array configurations without requiring modification of the underlying code.

To support deployment beyond a MATLAB development environment, standalone executables were generated using the MATLAB Compiler (Fig. 6). Precompiled binaries are provided for macOS and Windows (x64), enabling use of the optimisation tool on a standard laboratory computer without a local MATLAB installation. All versions share identical optimisation logic and numerical behaviour.

This packaging provides an accessible pathway for researchers to design, evaluate, and compare microphone array geometries tailored to specific localisation tasks.

3 RESULTS

Across all microphone counts and initialisation strategies, the optimisation procedure produced rapid early improvements in localisation performance, followed by clear saturation at configuration-specific pass-rate levels (Fig. 1). Pass-rate trajectories exhibited a consistent two-phase structure (Fig. 1c): an initial growth phase, in which performance increased steeply within the first 5–10 iterations, and a subsequent plateau phase, in which further coordinate updates yielded little or no additional improvement.

The early growth phase reflects the optimiser's ability to efficiently exploit the most informative geometric degrees of freedom. In contrast, the plateau phase indicates the presence of structural limits beyond which further refinement cannot substantially increase the fraction of test locations meeting the accuracy criterion. Importantly, this saturation behaviour was highly reproducible across repeated runs and was observed for both canonical geometric initialisations and random starts, suggesting that convergence is governed by geometry-dependent feasibility rather than by optimisation instability or insufficient runtime.

Representative visualisations illustrate how optimisation reshapes spatial error structure,

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eliminating geometry-specific failure modes (Fig. 1a). Large contiguous regions of high error are reduced after optimisation. These starting residual structures differ systematically between geometries, but the optimisation almost entirely eliminates them.

The distribution of per-microphone coordinate changes further demonstrates that optimisation operates primarily through modest adjustments rather than radical rearrangements (Fig. 1b). Typical displacements were on the order of ± 0.1 m per axis, with no evidence for large-scale geometric reconfiguration. Together, these trajectory-level results show that optimisation is effective at rapidly improving performance, and that even denser arrays, although starting at a high pass rate, can benefit from configurational optimisation.

3.1 Geometry-dependent limits on achievable pass rate

Optimisation outcomes differed systematically across array geometries, with configurations converging to distinct geometry-specific ceilings despite identical optimisation settings and repeated runs (Fig. 2). The distribution of run-wise maximum pass rates (Fig. 2, left) reveals clear performance stratification by array topology. Compact and highly symmetric configurations, the tetrahedral array, consistently plateaued at substantially lower pass rates than more spatially distributed geometries, including pyramidal, antiprismatic, and higher-order polyhedral layouts.

Final pass rates closely tracked the corresponding run-wise maxima for most configurations (Fig. 2, centre), indicating that the optimiser generally converged reliably once a favourable region of the solution space was reached. Configurations exhibiting broader inter-run variability showed larger discrepancies between maximum and final performance, suggesting sensitivity to local optima rather than premature termination of the optimisation.

Convergence behaviour exhibited a clear dependence on configuration, but this dependence did not mirror achievable performance ceilings. The iteration at which 95% of the run-wise maximum pass rate was first reached (Fig. 2, right) decreased systematically across configurations, with higher-order and random initialisations typically converging in fewer iterations than lower-order, highly constrained geometries. Notably, some of the fastest-converging configurations nonetheless achieved similar maximum pass rates, whereas several high-performing geometries required more iterations to approach their respective ceilings. This dissociation indicates that rapid convergence reflects a more tightly constrained optimisation landscape rather than superior localisation capability. Initial pass-rate discrepancies therefore arise primarily from geometry-imposed limits on the feasible solution space, not from differences in optimisation efficiency or termination behaviour.

Across pooled runs, moderate correlations were observed between maximum achievable pass rate and optimisation dynamics. The maximum pass rate correlated positively with the mean optimisation step size ($r = 0.64$), and negatively with overall root-mean-square microphone displacement ($r = -0.53$). These relationships indicate that higher-performing configurations tend to improve through coherent, moderately sized adjustments rather than through large cumulative geometric rearrangements. Importantly, extensive geometric drift was not predictive of superior performance. Together, these findings confirm that optimisation cannot compensate for topologically disadvantaged array designs; geometry selection therefore sets the fundamental limits on achievable localisation reliability.

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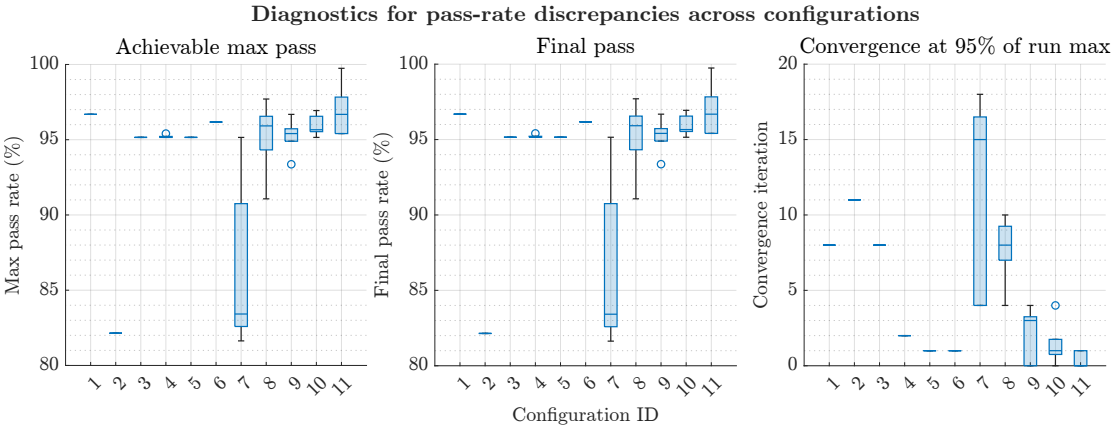


Figure 2: Runtime diagnostics underlying pass-rate discrepancies across microphone configurations. (Left) Distribution of the *achievable maximum pass rate* attained during optimisation, summarised across independent runs for each configuration. (Centre) *Final pass rate* at the end of optimisation, highlighting cases where convergence terminates below the run-wise maximum. (Right) *Convergence iteration*, defined as the iteration at which 95% of the run-wise maximum pass rate is first reached. Configuration IDs correspond to standard geometries (1–6) and random initialisations (7–11), as defined in Fig. 1b.

3.2 Spatial structure of localisation accuracy

Optimisation improved localisation accuracy across most of the workspace, but systematic spatial structure remained evident after convergence (Fig. 3). For all configurations, positional error increased with radial distance from the array centre, reflecting persistent geometric dilution of precision. Optimisation primarily reduced errors within a geometry-dependent sensing volume rather than eliminating distance-dependent degradation.

Positional accuracy showed strong dependence on array geometry (Fig. 3a). Compact and low-order configurations exceeded the positional acceptance threshold near the edge (∼1.9 m), but the median remained within the threshold, whereas more spatially distributed arrays maintained acceptable accuracy. These differences define geometry-specific effective sensing radii that optimisation alone cannot exceed, constrained by array topology and aperture — a design consideration important for optimising far-field accuracy.

Angular error exhibited a similar radial trend but weaker sensitivity to geometry (Fig. 3b). However, angular accuracy generally improved and remained stable (median <1 deg) as the microphone count increased. This indicates that positional accuracy is more strongly constrained by array geometry under the present optimisation objective.

Across all configurations, the 10–90% quantile bands for positional accuracy widened substantially at larger radii, revealing pronounced spatial heterogeneity in localisation reliability. Even after optimisation, peripheral regions of the workspace contained localised pockets of high error that could not be eliminated solely through coordinate refinement. However, the pass rate threshold was limited to 95; therefore, a more robust optimisation outcome can not be excluded. These results demonstrate that array geometry determines not only average accuracy but also the spatial extent over which reliable localisation can be achieved, and that these values can be optimised.

3.3 Optimisation reshapes error distributions

Baseline localisation performance varied widely across array geometries prior to optimisation, with large differences not only in pass rate but also in the structure of error distributions (Fig. 4).

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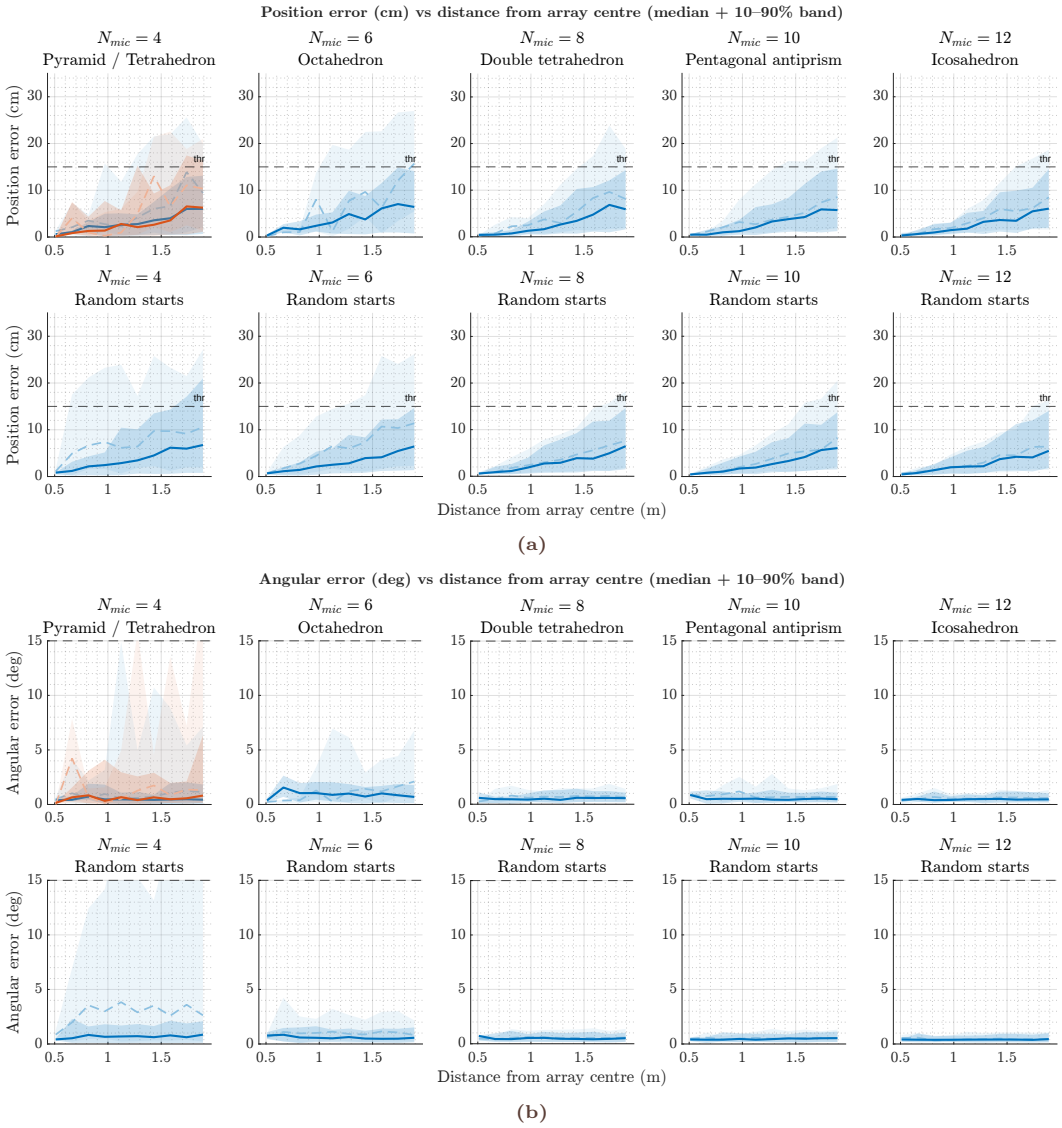


Figure 3: Optimisation improves spatial accuracy across distance. Performance is shown as median localisation error with a 10–90% band as a function of distance from the array centre, evaluated after optimisation. **(A)** Position error (cm) across standard geometric initialisations (top panels) and random starts (bottom panels) for increasing microphone counts ($N_{\text{mic}} = 4$–12). While optimisation reduces error across most of the workspace, residual performance differences persist between geometries, particularly at larger radii. (*Orange - tetrahedron*) **(B)** Angular error (deg) exhibits a similar distance-dependent structure but shows weaker sensitivity to geometry than position error. In both metrics, performance degrades systematically with distance, indicating that optimisation improves accuracy primarily within a geometry-constrained sensing volume rather than eliminating spatial limits entirely. Dashed lines indicate the predefined acceptance threshold used to compute pass rates in subsequent analyses.

In several configurations—particularly random initialisations and low-order canonical geometries—baseline localisation was characterised by heavy-tailed error distributions, with failures dominating the upper percentiles (see Appendix A for gain comparison tables).

Optimisation consistently reshaped these distributions by strongly suppressing upper-tail errors rather than uniformly reducing central tendency (Fig. 5). Across all geometries, reductions in median positional error were comparatively modest, typically on the order of a few centimetres. For example, median position error decreased from 11.6 cm to 3.8 cm for the pyramidal configuration ($N = 4$), from 13.0 cm to 4.2 cm for the tetrahedron, and from approximately 5–6 cm to 3–4 cm for higher-order polyhedral arrays. Similar modest shifts were observed for

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random initialisations once degenerate configurations were resolved.

In contrast, reductions in extreme positional error were substantial. The 95th percentile position error decreased by an order of magnitude in several cases, falling from values exceeding 1 m in poorly conditioned baseline geometries to ∼14–27 cm after optimisation. The most dramatic improvements occurred for random four-microphone arrays, where the 95th percentile positional error was reduced from approximately 5.9 m to 26 cm. Even for structured geometries with moderate baseline performance, such as the tetrahedron, the 95th percentile error dropped from 118 cm to 104 cm (see Appendix A), while higher-order configurations converged to consistently low upper-tail errors near the imposed threshold. These large reductions in tail error account for the majority of the observed gains in volumetric pass rate.

Angular error exhibited a qualitatively similar but weaker response to optimisation. Median angular error decreased modestly across most configurations, typically from ∼0.6–1.5° to ∼0.4–0.6°. Upper-tail angular errors were reduced substantially in configurations with poor baseline geometry—most notably random initialisations—where the 95th percentile angular error fell from tens or hundreds of degrees to values below 3°. However, for several canonical geometries, angular error distributions changed little after optimisation, and in some cases interquartile angular error increased slightly despite improved positional pass rates. Notably, for all configurations, the optimised medians fell below an average of 0.5 degrees.

This asymmetric response reflects the structure of the optimisation objective, which prioritises meeting a positional accuracy threshold with only a weak penalty on mean error. Optimisation therefore reallocates sensing precision toward eliminating catastrophic positional failures near the pass boundary, rather than uniformly improving accuracy across the entire workspace. As a result, volumetric reliability improves substantially even when median errors change only modestly.

Together, these results demonstrate that the principal effect of optimisation is the suppression of rare but severe localisation failures that dominate volumetric performance metrics. While this strategy is highly effective for maximising usable sensing volume, it also highlights a fundamental trade-off: objective functions tuned for threshold-based reliability may not optimise angular accuracy or spatial uniformity. Alternative formulations with carefully chosen parameters may therefore be required when applications demand consistent precision rather than robust pass-rate performance.

3.4 Microphone count effects and diminishing returns

Increasing microphone count improved achievable volumetric pass rates, but gains remained sublinear and exhibited clear diminishing returns under fixed-aperture constraints. While higher-N arrays tended to reach higher absolute pass rates, the incremental benefit per additional microphone decreased rapidly beyond low-order configurations, consistent with the economy analysis (Table 3).

To quantify whether localisation performance or optimisation behaviour scales systematically with microphone count, I evaluated a power-law model of the form

$$y = a\,N^{b}, \tag{9}$$

where N denotes the number of microphones and y is a run-wise summary metric. Fits were

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Table 1: Power-law scaling of optimisation behaviour with microphone count. Each metric y was fitted with a model $y = a\,N^b$ using log–log regression. Confidence intervals (CI) for a and b were estimated via bootstrap resampling ($B = 2000$). R^2 is reported in log space.

Metric	n	a	b	$a_{\mathrm{CI,lo}}$	$a_{\mathrm{CI,hi}}$	$b_{\mathrm{CI,lo}}$	$b_{\mathrm{CI,hi}}$	$R^2_{\log}$
Max. pass rate	25	77.5	0.095	70.3	88.3	0.038	0.139	0.514
Final pass rate	25	77.5	0.095	70.3	88.4	0.038	0.138	0.514
Convergence iter.	19	220	−2.10	51.3	1085	−2.87	−1.36	0.683
Mean step size	25	0.0105	0.195	0.0052	0.0222	−0.16	0.53	0.032
Drift RMS	25	0.681	−1.31	0.358	1.31	−1.63	−1.02	0.770

performed in log–log space using ordinary least squares, with uncertainty estimated via bootstrap resampling (2000 samples; 95% confidence intervals). The analysis was restricted to the random-start N-sweep ($N = 4$–12) to isolate the effect of microphone count from specific geometric symmetries.

As summarised in Table 1, both maximum and final volumetric pass rates exhibit a weak but positive scaling with microphone count ($b \approx 0.095$), explaining approximately half of the variance in log space ($R^2 \approx 0.51$). While this indicates that higher-order arrays can, on average, achieve higher pass rates, the shallow exponent confirms that these gains increase slowly with N and fall far short of proportional scaling. In practical terms, doubling the number of microphones yields only marginal improvements in usable localisation volume.

In contrast, optimisation dynamics exhibit stronger, more systematic scaling with microphone count. The iteration at which the optimiser converges decreases steeply with N ($b \approx -2.10$), indicating that higher-order arrays saturate the objective function more rapidly. Geometry drift also decreases markedly with increasing N ($b \approx -1.31$, $R^2 = 0.77$), reflecting increased geometric redundancy and reduced sensitivity to individual microphone perturbations. These trends suggest that additional microphones primarily stabilise the optimisation landscape rather than enabling proportionally better localisation performance.

Mean optimisation step size shows no reliable dependence on microphone count (CI spanning zero; $R^2 \approx 0.03$), indicating that the local move scale accepted by the optimiser is governed by geometric conditioning rather than sensor count alone.

Taken together, these results demonstrate a clear decoupling between optimisation behaviour and task-level performance. While higher microphone counts constrain the search space and promote rapid, stable convergence, they do not confer commensurate gains in volumetric localisation reliability. This finding reinforces the central conclusion of this study: carefully optimised low-order arrays can achieve high and robust localisation performance with substantially greater sensor economy than denser configurations.

3.5 Optimisation effects on array compactness

To assess whether improvements in localisation performance required physically larger arrays, I quantified the axis-aligned bounding-box dimensions of each configuration before and after optimisation (Table 2). For each run, array width, height, and depth were computed from microphone coordinates, and the maximum span across axes was used as a conservative measure of overall array extent.

Across canonical geometries, optimisation generally resulted in moderate expansions of array size, though the magnitude and direction of change depended strongly on topology. Median max-span ratios ranged from 1.11 (pentagonal antiprism, 10 microphones) to 1.85 (octahedron,

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Table 2: Array compactness before and after optimisation. For each configuration (five runs each), median axis-aligned bounding-box spans (width W, height H, depth D; metres) are reported from microphone coordinates before $(\cdot_0)$ and after $(\cdot_1)$ optimisation. The maximum span max is the largest of (W, H, D) for a given state and provides a conservative measure of overall array extent. The ratio $\max_1 / \max_0$ indicates whether optimisation expanded (> 1) or compacted (< 1) the array.

ID	Configuration	n	W_0	H_0	D_0	W_1	H_1	D_1	$\max_0$	$\max_1$	$\frac{\max_1}{\max_0}$
1	Pyramid (N_{mic}=4)	5	0.500	0.500	0.400	0.478	0.625	0.557	0.500	0.625	1.251
2	Tetrahedron (4)	5	0.491	0.491	0.491	0.620	0.707	0.529	0.491	0.707	1.441
3	Octahedron (6)	5	0.500	0.500	0.500	0.922	0.340	0.498	0.500	0.922	1.845
4	Double tetrahedron (8)	5	0.612	0.707	0.750	0.693	0.792	0.838	0.750	0.838	1.118
5	Pentagonal antiprism (10)	5	0.637	0.607	0.318	0.599	0.708	0.371	0.637	0.708	1.111
6	Icosahedron (12)	5	0.638	0.638	0.638	0.638	0.638	0.763	0.638	0.763	1.196
7	Random (4)	5	0.517	0.329	0.380	0.663	0.485	0.398	0.517	0.663	1.283
8	Random (6)	5	0.388	0.474	0.541	0.457	0.657	0.568	0.541	0.657	1.215
9	Random (8)	5	0.535	0.488	0.554	0.551	0.632	0.642	0.554	0.642	1.159
10	Random (10)	5	0.568	0.598	0.619	0.568	0.654	0.696	0.619	0.696	1.125
11	Random (12)	5	0.646	0.668	0.523	0.646	0.675	0.528	0.668	0.675	1.011

6 microphones), corresponding to absolute increases of approximately 7–42 cm. The pyramidal and tetrahedral four-microphone configurations exhibited substantial expansion (ratios of 1.25 and 1.44, respectively), reflecting the need to escape geometrically constrained initial layouts. In contrast, higher-order canonical arrays showed comparatively restrained growth: the icosahedral configuration (12 microphones) increased only modestly (ratio = 1.20), despite its higher sensor count.

Random initialisations showed larger, more variable changes in compactness, particularly at low microphone counts. The random four-microphone configuration showed a median max-span ratio of 1.28, indicating pronounced expansion from initially compact or near-degenerate geometries. However, as the microphone count increased, random arrays exhibited progressively smaller relative changes, with the random 12-microphone configuration remaining almost unchanged after optimisation (ratio = 1.01). This pattern indicates that higher-order arrays inherently provide sufficient geometric flexibility to accommodate optimisation without substantial physical expansion.

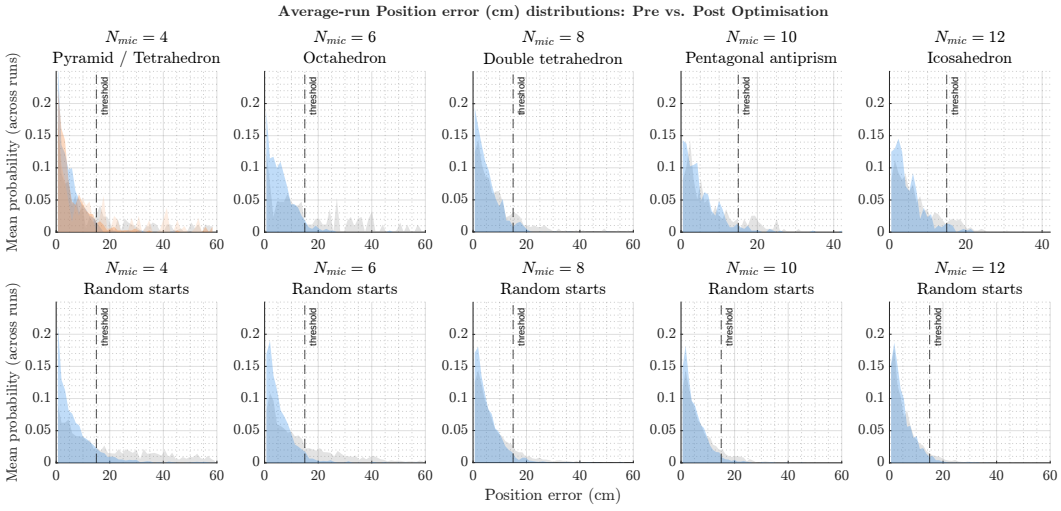
Importantly, no monotonic relationship was observed between microphone count and final array size. Optimised arrays with 8–12 microphones occupied comparable spatial extents despite differing sensor counts, reinforcing the conclusion that improved performance at higher N arises primarily from geometric redundancy rather than increased physical footprint. These results demonstrate that substantial gains in volumetric localisation reliability can be achieved without proportional increases in array size, supporting the feasibility of compact, portable array designs for field deployment.

4 DISCUSSION

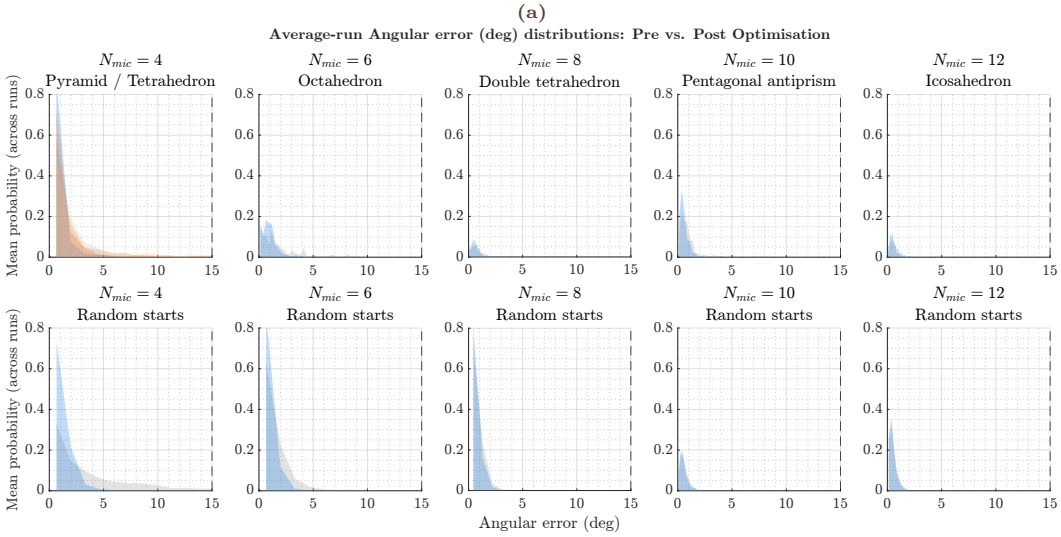
This study introduces a task-oriented framework for microphone-array optimisation that explicitly links array geometry to volumetric localisation reliability under realistic field constraints. The central finding challenges a common assumption in acoustic array design: increasing microphone count does not guarantee proportional improvements in localisation performance. Instead, carefully optimised low-order arrays—particularly four-to-six-microphone configurations—can achieve comparable or superior performance to higher-order systems when evaluated holistically through metrics of economy and responsiveness to optimisation.

This result has direct practical relevance for bioacoustic field studies. In deployments where

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(a)



(b)

Figure 4: Localisation performance across array geometries. Average-run error distributions are shown for positional (a) and angular (b) localisation accuracy across microphone-array configurations and random initialisations. Distributions are pooled across runs and source locations for each configuration. The vertical dashed line indicates the task-level accuracy threshold for successful localisation. Substantial performance differences are already present at baseline, with some geometries yielding a larger fraction of low-error solutions than others, motivating the optimisation analyses that follow.

portability, setup speed, and mechanical robustness are paramount, optimised four-microphone arrays offer a favourable balance between accuracy and deployability. When normalised by sensor count, four-microphone configurations consistently exhibit the highest performance gain per microphone, making them particularly efficient for near to mid-field volumetric localisation. These findings motivate a re-evaluation of common array-design heuristics in bat bioacoustics, where larger arrays are often adopted by default without systematic assessment of whether their additional complexity yields proportional gains in usable sensing volume.

More broadly, the results highlight a persistent yet under-addressed issue in acoustic tracking studies: although three-dimensional localisation of freely flying bats has been reported in numerous investigations, the quantitative characterisation of localisation accuracy is rarely discussed in detail. In many cases, localisation serves as a means to downstream behavioural interpre-

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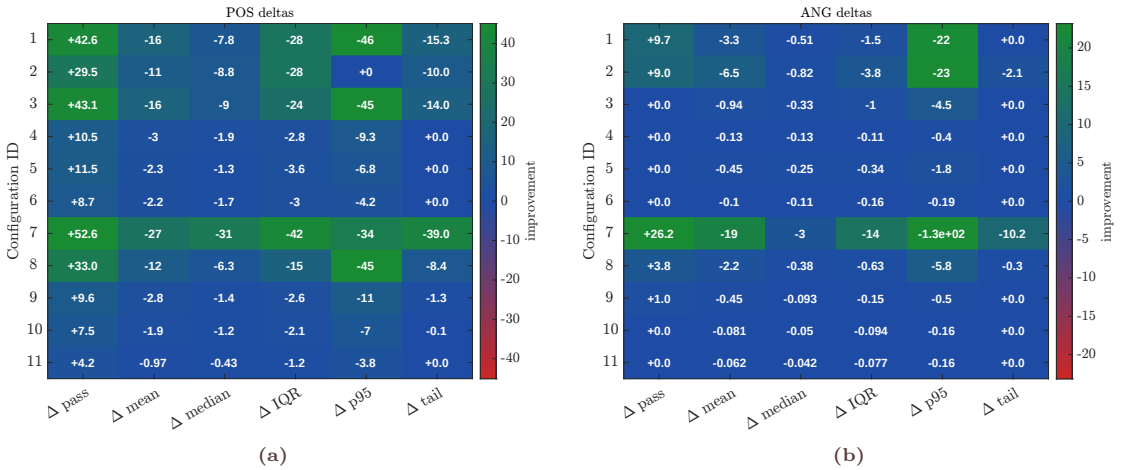


Figure 5: Optimisation-induced changes in localisation error. For each optimisation run, errors were evaluated on the same test grid at the first recorded iteration ($it = 0$; *before*) and at the selected *after* state (final iteration in the current analysis). Per-grid-point differences were computed as $\Delta e = e_{\text{after}} - e_{\text{before}}$ using the positional error (a) and the angular error (b). The differences were summarised using robust statistics, including the 95th percentile error (p_{95}) to characterise tail behaviour. Distributions of Δe are shown by configuration

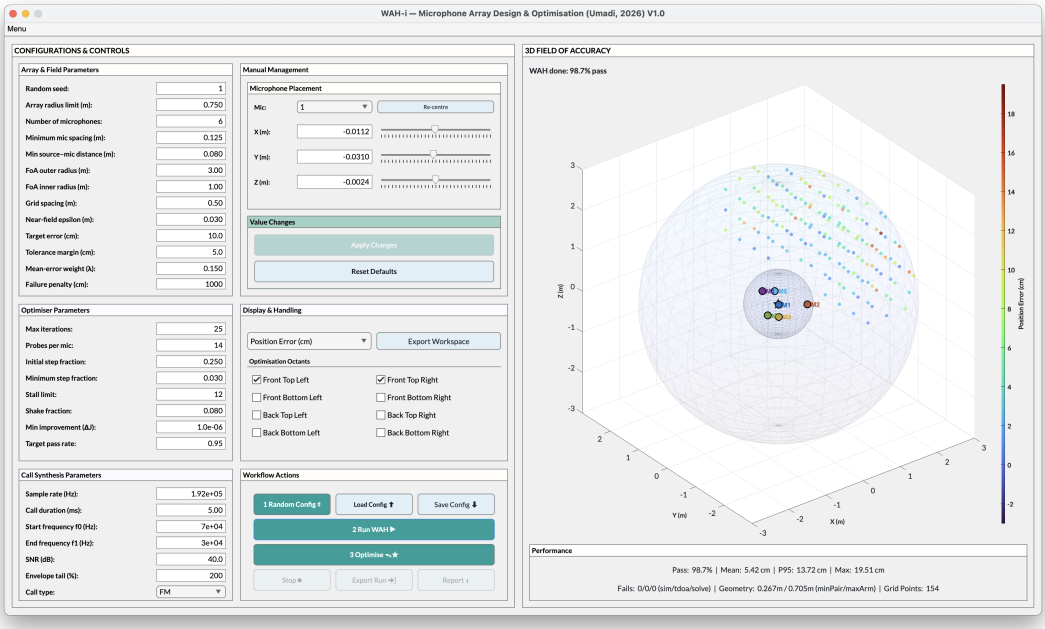


Figure 6: Interactive interface for array optimisation and performance exploration (WAH-*i* V1.0). The application provides an accessible entry point for exploring microphone array configurations under user-defined accuracy and spatial constraints. Users can specify the number of microphones, select or initialise array geometries, define aperture limits and localisation accuracy thresholds, and run the optimisation interactively. The interface visualises optimisation progress, the array geometry through optimisation, and the spatial distribution of localisation error within the evaluation volume.

tation (e.g., [7, 13, 15, 48, 49]) rather than as an explicitly evaluated measurement outcome. As a result, uncertainties in spatial accuracy can remain implicit, providing only circumstantial support for behavioural hypotheses.

The implications of such uncertainties are non-trivial. For example, a positional error of 0.25–0.5 m over a 50 ms interval—comparable to a typical inter-pulse interval during search or approach—corresponds to a velocity uncertainty of approximately ±5–10 $\mathrm{m\,s^{-1}}$. This range is on

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the order of the mean flight speed of vespertilionid bats such as *Pipistrellus* and *Myotis*, implying that biologically meaningful kinematic signatures can be obscured by localisation error alone. Extreme errors may be identifiable and excluded—for instance, an apparent change in velocity from 2 $\mathrm{m\,s^{-1}}$ to 20 $\mathrm{m\,s^{-1}}$ over 50 ms implies an acceleration exceeding 36 g, which is physiologically implausible. However, more moderate inaccuracies can remain hidden, manifesting as plausible yet misleading flight trajectories.

Furthermore, raw three-dimensional localisation coordinates, when visualised as flight paths or smoothed for statistical analysis, can produce convincing reconstructions even when underlying uncertainties are substantial. Although many studies mitigate these issues by restricting analysis volumes or excluding outliers, the absence of a principled method for optimising array geometry leaves open the question of whether the chosen configuration was ever capable of resolving the behavioural features under investigation. In this context, the present optimisation framework directly addresses a foundational limitation in acoustic tracking: it provides a systematic means to design and evaluate microphone arrays based on task-specific accuracy requirements, rather than relying on heuristic or legacy configurations.

This study extends the Widefield Acoustics Heuristic [42] by inverting its original forward perspective. Whereas the earlier framework characterised the accuracy produced by a given array geometry, the present work uses that characterisation as an objective function to iteratively refine the geometry to achieve a desired field of accuracy. In this sense, the approach transforms array design from a descriptive exercise into a constructive one.

The optimisation and evaluation were restricted to forward-facing upper octants, reflecting typical field deployments where arrays are oriented toward expected flight paths rather than enclosing the source volume isotropically. This restriction improves ecological relevance while reducing computational cost. Given a grid resolution of 0.25 m, 55 optimisation trajectories, 25 iterations per run, and 11 configurations, expanding the evaluated volume would substantially increase computational expense without altering the underlying optimisation logic. Importantly, the algorithm itself is agnostic to the chosen field of audition: with sufficient computational budget, the same framework can be applied to alternative spatial constraints, target volumes, or task-specific accuracy thresholds.

The optimisation problem addressed here is inherently geometric. Localisation error in TDOA-based systems arises from the interaction between timing uncertainty and sensor geometry, commonly expressed through geometric dilution of precision (GDOP) [31, 35]. Even under ideal timing conditions, array geometry imposes structural limits on the regions of space that can be resolved accurately. Crucially, these limits are highly non-uniform and depend on both array topology and source position.

The results demonstrate that while higher microphone counts improve raw spatial sampling density, the gains saturate rapidly once geometric constraints dominate error structure (Figs. 2 & 3). The observed convergence behaviour (Fig. 1) supports the interpretation that optimisation reveals intrinsic performance limits. Adding microphones within a fixed aperture does not fundamentally overcome distance-dependent degradation in localisation accuracy (Fig. 3), reinforcing the notion that geometry, rather than sensor count alone, governs usable sensing volume.

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Table 3: Economy–responsiveness comparison of array geometries. Performance gain is defined as the absolute increase in volumetric pass rate following optimisation. Economy is quantified by the number of microphones N, and responsiveness is measured as gain per microphone ($\Delta\text{pass}/N$). Arrays are ranked primarily by positional economy (rank_{pos}), with the angular economy shown for comparison. Higher responsiveness indicates a larger improvement in volumetric localisation reliability per sensor, highlighting configurations that benefit most from optimisation under fixed microphone budgets.

ID	Geometry	N	Δ_{pos}	Δ_{pos}/N	rank_{pos}	Δ_{ang}	Δ_{ang}/N	rank_{ang}
1	Random	4	52.60	13.15	1	26.17	6.54	1
2	Pyramid	4	42.60	10.65	2	9.69	2.42	2
3	Tetrahedron	4	29.46	7.36	3	8.97	2.24	3
4	Octahedron	6	43.11	7.19	4	0.00	0.00	8
5	Random	6	32.96	5.49	5	3.83	0.64	4
6	Double tetrahedron	8	10.51	1.31	6	0.00	0.00	6
7	Random	8	9.59	1.20	7	1.02	0.13	5
8	Pentagonal antiprism	10	11.48	1.15	8	0.00	0.00	9
9	Random	10	7.50	0.75	9	0.00	0.00	10
10	Icosahedron	12	8.67	0.72	10	0.00	0.00	7
11	Random	12	4.23	0.35	11	0.00	0.00	11

4.1 Geometric comparison of gained performance

To disentangle geometric constraints from optimisation artefacts, I evaluated canonical three-dimensional geometries — tetrahedral, pyramidal, octahedral, antiprismatic, and icosahedral — alongside randomly initialised arrays with matched microphone counts. Canonical geometries are attractive in practice because they are mechanically simple to construct, often benefit from symmetry, and can be realised using standard structural components such as commercially available joints and connectors. However, such geometries could, in principle, act as restrictive initialisation points that bias optimisation toward local optima.

Random initialisations address this concern by sampling the configuration space stochastically, providing a broader representation of possible geometries. Because the space of valid microphone configurations is effectively continuous and infinite, both canonical and random configurations should be interpreted as sparse samples from an underlying high-dimensional design space. Repeated optimisation from both structured and random initial conditions allows separation of geometry-imposed performance ceilings from optimisation path-dependence (Fig. 2). The convergence of both initialisation strategies toward similar performance levels provides strong evidence that observed limits are intrinsic to geometry rather than artefacts of algorithm design.

4.2 Microphone economy and performance scaling

The economy–responsiveness analysis provides a quantitative basis for ranking array geometries in terms of performance gain per microphone (Table 3). Across both canonical and random configurations, four-microphone arrays consistently rank highest in positional economy, achieving the largest absolute increases in volumetric pass rate when normalised by sensor count. While higher-order arrays often achieve slightly higher absolute pass rates, the marginal benefit per additional microphone decreases sharply beyond $N = 4 - 6$.

This pattern is reinforced by the power-law scaling analysis (Table 1). Neither the maximum nor the final volumetric pass rate shows a meaningful dependence on microphone count, with scaling exponents close to zero and confidence intervals that span zero. In contrast, optimisation dynamics show strong and systematic scaling with N: the convergence iteration count and overall geometric drift both decrease approximately as N^{-2} and N^{-1}, respectively. These results indicate that increasing array order primarily constrains the optimiser's degrees of freedom,

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rather than yielding proportional improvements in usable localisation volume.

At the same time, denser arrays confer increased robustness to structural perturbations. With more microphones, small deviations in individual sensor positions exert a reduced influence on the global solution, effectively smoothing the error landscape. Under conditions where additional geometric freedom is available—such as larger allowable aperture sizes, extended arm lengths, or actively adjustable mounting geometries—higher-order arrays may therefore access substantially improved accuracy profiles [50, 51]. In such cases, the benefits of increased redundancy and spatial sampling may outweigh the diminishing returns, highlighting that the optimal microphone count is ultimately task- and platform-dependent.

In field settings where the primary objective is reliable near to mid-field localisation—such as reconstructing flight trajectories, estimating spatial decision rules, or resolving interactions among nearby individuals—optimised four-microphone arrays therefore offer a favourable balance between accuracy, responsiveness to optimisation, and economy. The modest and saturating gains achieved by denser arrays suggest that additional sensors are often better traded for improvements in timing precision, deployment geometry, or experimental flexibility, rather than increased microphone count alone.

4.3 Accuracy: task-specific design considerations

Positional accuracy was treated as the primary optimisation objective, reflecting its central role in reconstructing three-dimensional trajectories in the near field. Angular accuracy was evaluated as a secondary metric, providing complementary information about directional resolution. As expected, angular accuracy improved with increasing microphone count (Table 3), reflecting the increased directional diversity afforded by denser arrays.

However, the economic analysis reveals that angular gains per microphone remain lower for higher-order configurations. Consequently, the choice of microphone count should be guided by task requirements: applications prioritising precise directional cues (e.g., beamforming) may justify denser arrays, whereas studies focused on spatial reconstruction within a limited volume benefit more from optimised low-order geometries [52]. The angular error distributions shown in Figures 3 and 4 further demonstrate that denser arrays yield more robust and consistently low angular errors across distance, making them particularly well suited for far-field applications such as source separation and directional filtering [50, 51], where angular stability rather than absolute positional accuracy is the primary constraint.

Beyond localisation accuracy, optimised microphone-array geometry has direct implications for acoustic-source separation in conservation bioacoustics, where emerging techniques are actively addressing the problem at individual to population, and niche levels [18, 19, 21, 53, 54]. In dense acoustic environments, overlapping calls from multiple individuals or species constitute a major limitation for abundance estimation and behavioural inference. Arrays that yield poorly resolved spatial cues yield ambiguous direction estimates and unstable separation, even at high signal-to-noise ratios. By maximising volumetric localisation reliability under realistic constraints, the proposed optimisation framework implicitly improves the spatial separability of acoustic sources throughout the observation volume. This establishes a natural extension of the method toward array designs tailored not only for localisation, but also for multi-source disentanglement in conservation monitoring contexts, thus directly contributing to ongoing developments [18, 55, 56].

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4.4 Build and deployment considerations

An important practical insight is that relatively small geometric perturbations can produce measurable changes in localisation performance (Fig. 5). This sensitivity implies that unintentional deviations during construction or deployment—such as slight arm misalignments or mounting asymmetries—can materially affect array performance. Consequently, careful attention to mechanical design and reproducibility is essential, particularly for arrays operating near the threshold of acceptable accuracy.

At the same time, the finding that optimised four-microphone arrays achieve very high performance opens new possibilities for reusable, adaptable array systems. Compact hardware platforms such as Batsy4-Pro [23] can be paired with adjustable or telescopic arm structures mounted on ball joints, allowing rapid reconfiguration for task-specific optimisation. This combination enables a single system to be repurposed across laboratory and field contexts, enhancing both logistical efficiency and experimental control.

4.5 Extended applications

Beyond immediate design guidance, the optimisation runs presented here generate a rich, structured dataset linking array geometry, optimisation trajectories, and performance metrics. Such data provide a natural substrate for data-driven models that can learn mappings between geometric features and localisation reliability, convergence behaviour, or robustness under limits. Supervised or surrogate modelling [57] approaches—ranging from regression models to lightweight neural networks [58]—could be trained on these optimisation histories to rapidly predict promising regions of configuration space, thereby reducing the need for exhaustive iterative evaluation [59]. In this way, optimisation-derived datasets offer a pathway toward hybrid model-based and learning-based array design, where computationally expensive physics-based simulations [60] are used sparingly to refine solutions proposed by learned priors, substantially lowering the cost of exploring high-dimensional array geometries.

In addition to static optimisation, the proposed framework naturally extends to the design and control of *adaptive* or *reconfigurable* microphone arrays. Because the optimisation objective is defined directly in terms of task-level localisation performance, it can be re-evaluated as array geometry evolves, enabling closed-loop adjustment of sensor positions in response to changing acoustic or environmental conditions. When combined with mechanically adjustable mounts or articulated array elements, this allows microphone configurations to be updated dynamically to maintain a favourable localisation geometry as sound sources move through the observation volume or as the region of interest shifts.

Adaptive optimisation becomes particularly powerful when coupled with auxiliary sensing modalities. Inertial sensors, visual tracking, radar, or coarse depth estimation can provide approximate source trajectories or scene constraints, which can be used to bias the optimiser toward geometries that maximise localisation accuracy in the predicted source region. For sources with known or stereotyped emission characteristics—such as echolocating bats, calling amphibians, or vocalising birds—the optimiser can further exploit prior knowledge of signal bandwidth, directionality, or temporal structure to favour configurations that enhance spatial discrimination for those signals specifically.

Beyond bioacoustic and robotic sensing applications, the WAH-i framework also has po-

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tential relevance for the design of next-generation hearing-assistive devices. Modern hearing aids increasingly rely on multi-microphone arrays to enhance spatial selectivity, suppress interference, and support directional tracking of a target speaker in complex acoustic scenes [61, 62]. Although hearing aids operate at vastly different spatial scales than the arrays considered here, the underlying optimisation principles—maximising directional discrimination, robustness, and coverage given limited sensor count—are directly transferable [63]. As research into spatial hearing, beamforming, and active speaker tracking in hearing aids continues to advance, incorporating machine learning approaches [64, 65], optimisation-driven design frameworks such as WAH-i may provide a useful tool for identifying geometries that balance perceptual benefit against size, power, and mechanical constraints.

More generally, this perspective reframes array geometry as a *dynamic state variable* rather than a fixed design choice. By integrating geometric optimisation with real-time sensing and tracking, microphone arrays can be made responsive to both source behaviour and task demands, enabling sustained localisation performance under motion and clutter. The present framework, therefore, provides a principled foundation for future adaptive array systems that unify sensing, optimisation, and actuation within a single performance-driven control loop.

Limitations

The present analysis considers only frequency-modulated (FM) calls emitted by a stationary source—the tool provides provisions for the CF call type. In principle, stationary constant-frequency (CF) calls would yield comparable localisation performance, as the underlying time-difference-of-arrival (TDOA) assumptions remain unchanged. However, once source motion is introduced, FM and CF calls diverge markedly in their localisation behaviour. Motion-induced effects disproportionately affect CF calls, which are highly susceptible to systematic localisation errors under even modest relative velocities [42], whereas short FM chirps retain robust median accuracy despite velocity-dependent broadening of their error distributions. This asymmetry has been demonstrated explicitly in simulations using the Array Widefield Acoustics Heuristic, where increasing source speed primarily inflates the upper tail of the error distribution rather than collapsing localisation performance entirely [23].

All simulations in the present study assume idealised acoustic propagation, including homogeneous sound speed, absence of multipath reflections, and perfect sensor synchrony [42, 43]. These simplifications are deliberate: they allow isolation of geometry-dependent effects and provide a controlled baseline against which additional stochastic or environmental factors can be considered post hoc. Incorporating atmospheric gradients, scattering, or clutter would substantially increase model complexity without altering the fundamental geometric constraints explored here.

The optimisation framework is computationally demanding, as each iteration requires full volumetric evaluation of localisation performance. Consequently, users should exercise caution when running large optimisation batches on machines with limited processing resources.

Finally, the optimisation algorithm explores microphone positions within local bounds around an initial configuration. Although this strategy reliably reveals geometry-dependent performance ceilings, it does not guarantee discovery of a global optimum. Broader exploration can be achieved by relaxing step-size constraints or increasing perturbation scales, but this comes at the expense of substantially increased computational cost. In practice, the results indicate

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that topology selection dominates performance, and that optimisation primarily serves to refine, rather than fundamentally transform, a given array geometry.

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COMPETING INTERESTS

The author declares no competing interests.

ETHICAL STATEMENT

This study did not involve animal handling, experimentation, or data collection from live animals. All analyses were conducted using simulated data and previously established acoustic models; therefore, no ethical approval was required.

DATA & CODE AVAILABILITY

All data and code used in this study, as well as the WAH-*i* installable packages, are available on GitHub at `https://github.com/raviumadi/WAHi-Optimiser`.
User-guide and the documentation for the WAH-*i* software are published at `https://raviumadi.github.io/WAHi-Optimiser`.

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A PERFORMANCE GAIN TABLES

A.1 Positional Performance

Table 4: Positional error statistics before and after optimisation. Pass rate denotes the percentage of grid points with positional error ≤ 15 cm. Error statistics (mean, median, interquartile range (IQR), and 95th percentile (p95)) are reported in centimetres.

Configuration	N	Pass_0 (%)	Mean_0 (cm)	Median_0 (cm)	IQR_0 (cm)	p95_0 (cm)	Pass_1 (%)	Mean_1 (cm)	Median_1 (cm)	IQR_1 (cm)	p95_1 (cm)
Pyramid	4	54.1	32.7	11.6	34.6	136.8	96.7	5.1	3.8	6.1	13.6
Tetrahedron	4	52.7	31.0	13.0	38.0	118.3	82.1	15.9	4.2	9.6	104.1
Octahedron	6	52.0	93.1	13.8	30.5	84.9	95.2	6.0	4.8	6.5	15.0
Double tetrahedron	8	84.7	8.3	6.0	8.8	24.2	95.2	5.3	4.0	5.9	14.9
Pentagonal antiprism	10	83.7	7.6	5.2	9.1	21.5	95.2	5.3	3.8	5.5	14.8
Icosahedron	12	87.5	7.1	5.2	7.7	18.6	96.2	4.9	3.5	4.7	14.4
Random	4	33.9	2608.9	36.3	104.6	587.5	86.5	11.1	4.9	8.5	26.4
Random	6	62.3	518.4	10.4	20.6	99.0	95.3	5.7	4.0	5.8	14.9
Random	8	85.7	9.2	5.3	8.4	25.5	95.3	5.5	4.0	5.8	14.6
Random	10	88.5	7.1	5.0	7.6	21.2	96.0	5.2	3.8	5.5	14.2
Random	12	92.7	5.9	4.1	6.3	17.3	96.9	5.0	3.6	5.1	13.4

A.2 Angular Performance

Table 5: Angular localisation error statistics before and after optimisation. Pass rate denotes the percentage of grid points with angular error $\leq 15^\circ$. Angular error statistics (mean, median, interquartile range (IQR), and 95th percentile (p95)) are reported in degrees.

Configuration	N	Pass_0	Mean_0 (deg)	Median_0 (deg)	IQR_0 (deg)	p95_0 (deg)	Pass_1 (%)	Mean_1 (deg)	Median_1 (deg)	IQR_1 (deg)	p95_1 (deg)
Pyramid	4	90.3	4.02	0.97	2.14	24.6	100.0	0.72	0.45	0.59	2.26
Tetrahedron	4	82.1	13.4	1.46	5.24	75.7	91.1	6.90	0.64	1.39	52.5
Octahedron	6	100.0	1.91	1.22	1.79	6.78	100.0	0.98	0.88	0.77	2.32
Double tetrahedron	8	100.0	0.76	0.69	0.62	1.76	100.0	0.63	0.55	0.51	1.37
Pentagonal antiprism	10	100.0	1.04	0.72	0.79	3.24	100.0	0.59	0.47	0.45	1.48
Icosahedron	12	100.0	0.66	0.57	0.54	1.43	100.0	0.55	0.46	0.38	1.24
Random	4	73.8	20.2	3.63	15.2	136.2	99.9	0.98	0.68	1.14	2.69
Random	6	96.2	2.90	0.97	1.25	7.67	100.0	0.73	0.58	0.62	1.85
Random	8	98.9	1.05	0.58	0.62	1.85	99.9	0.60	0.49	0.47	1.35
Random	10	100.0	0.64	0.52	0.52	1.51	100.0	0.56	0.47	0.43	1.35
Random	12	100.0	0.54	0.44	0.45	1.31	100.0	0.48	0.39	0.37	1.15

B WAH-*I* RUNS IMPLEMENTATION

The following sections describe the implementation and analysis procedures with explicit reference to variable names and structures used in the accompanying codebase; all such variables are defined and documented within the released source code.

B.1 Batch experimental design and run generation

The batch runner (`run_iwah_experiment_batch.m`) executed optimisation runs under two start-condition classes:

1. **CSV-defined canonical geometries ("CSV starts").** Six standard geometries were provided as CSV files (`configs/*.csv`), each defining an initial array layout in Cartesian coordinates:

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- 4-microphone Pyramid (`4mics_Pyramid.csv`)
- 4-microphone Tetrahedron (`4mics_Tetrahedron_small.csv`)
- 6-microphone Octahedron (`6mics_Octahedron.csv`)
- 8-microphone Double tetrahedron (`8mics_double_tetrahedorn_small.csv`)
- 10-microphone Pentagonal antiprism (`10mics_pentagonal_antiprism.csv`)
- 12-microphone Icosahedron (`12mics_icosahedron.csv`)

If enabled (`EXP.useCsvNativeN=true`), the microphone count N_{mic} was read directly from the number of rows in the CSV file; otherwise, N_{mic} could be overwritten by the batch settings.

2. **Random-start geometries ("random starts").** For each $N_{\mathrm{mic}} \in \{4, 6, 8, 10, 12\}$, a random initial geometry was generated internally by the optimiser (see below) and then refined.

For each condition (canonical or random) I ran $K = 5$ independent repeats (`EXP.repeats=5`). Unless debugging, runs were executed in parallel using MATLAB's `parfor` (`EXP.useParfor=true`) with a local pool (`parpool`).

B.1.1 Simulation space and test grid

All optimisation runs were evaluated on a common three-dimensional localisation test grid defined by:

- outer radius $R = 2.0$ m (`P.R`)
- inner exclusion radius $R_{\mathrm{in}} = 0.30$ m (`P.Rin`)
- grid step $s = 0.250$ m (`P.step`)

The grid thus sampled candidate source locations in a volumetric shell around the array centre (origin). For each iteration, the optimiser computed localisation errors at these grid points (see below) and stored the grid points as `iter(k).gridPts`.

B.1.2 Geometric constraints and feasibility

To prevent degenerate geometries and to enforce practical build constraints, candidate microphone layouts were constrained as follows:

- **Bounded array extent:** all microphone positions were constrained within a sphere of radius $L_{\max} = 0.50$ m (`P.Lmax`).
- **Minimum inter-microphone distance:** microphone pairwise distances were constrained to exceed `P.minPairDist` (=6.25 cm).
- **Minimum source-to-microphone distance:** grid points too close to any microphone were excluded using `P.minDistToAnyMic`=0.08 m, with a soft exclusion parameter `P.epsDist`=0.03 m.

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For CSV-defined geometries, the batch pipeline recentered the array coordinates to the origin (`CSV.recenter=true`) and optionally repaired minor constraint violations (`CSV.autoRepair=true`). The repair procedure attempted up to 200 iterations (`CSV.repairMaxTries`) with a jitter perturbation that increased multiplicatively (`CSV.repairJitter0=1e-3`, growth `CSV.repairJitterGrowth=1.15`). If enabled, repair clamped extents to $L_{\max}$ (`CSV.repairClampLmax=true`) and preferentially applied scaling adjustments (`CSV.repairPreferScale=true`).

B.1.3 Acoustic signal and propagation parameters

Each localisation evaluation was based on a simulated biosonar-like call, parameterised by:

- sampling rate $f_s = 192$ kHz (`P.fs`)
- call duration $d = 5$ ms (`P.d`)
- FM sweep from $f_0 = 30$ kHz to $f_1 = 70$ kHz (`P.f0`, `P.f1`)
- additive noise level `P.snr_db`=40 dB
- tail window parameter `P.tail`=200 (used internally by the call generator/processing)
- call type `P.callType='FM'`
- source velocity $\mathbf{v} = [0, 0, 0]$ (`P.velocity`), i.e. stationary source
- speed of sound $c = 343\ \mathrm{m\,s^{-1}}$ (`P.c`)

These parameters were held constant across conditions and repeats to isolate the effect of geometry and microphone count.

B.1.4 Optimisation objective, pass criterion, and penalties

At each iteration, the optimiser produced localisation error statistics over the test grid and computed:

- **Position error** e_{pos} in cm (`stats.err_cm`), and/or
- **Angular error** e_{ang} in degrees (`stats.ang_deg`)

depending on the selected evaluation metric.

A grid point was deemed a *pass* if its position error satisfied

$$e_{\mathrm{pos}} \leq \theta, \tag{10}$$

where the pass threshold was defined as `P.threshold_cm` with

$$\theta = \texttt{P.best_cm} + \texttt{P.tol_cm} = 10 + 5 = 15\ \mathrm{cm}. \tag{11}$$

The *pass rate* (`stats.passRate`) was the fraction of grid points passing the criterion.

To discourage pathological failures, runs used a large penalty for failing or invalid points (`P.failPenalty_cm`=1000). Additionally, the optimiser targeted a desired pass rate `P.targetPass`=0.95 and used an internal weighting parameter `P.lambdaMean`=0.15 to balance mean error and pass-based objectives (exact internal objective combination is implemented within `iwah_run_optimiser_core` and recorded as `iter(k).J`).

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B.1.5 Optimiser configuration and stopping behaviour

Optimisation used the following control parameters (`OPT`):

- maximum iterations: 25 (`OPT.maxIters`)
- probing density: 14 probes per microphone per iteration (`OPT.probesPerMic`)
- initial step fraction: 0.25 (`OPT.initStepFrac`)
- minimum step fraction: 0.03 (`OPT.minStepFrac`)
- stall limit: 12 (`OPT.stallLimit`)
- "shake" fraction: 0.08 (`OPT.shakeFrac`)
- improvement tolerance: `OPT.improveEpsJ`=10^{-6}

The optimiser recorded intermediate shake steps using non-integer iteration indices (e.g. $it = 0.5$), which were later excluded from trajectory summaries to align runs on integer iteration counts.

B.1.6 Octant selection

The experiment allowed restricting evaluations to specific spatial octants (`Q`). In the current batch settings, only two regions were enabled: `frontTopLeft=true` and `frontTopRight=true`, while all others were disabled. This restriction was applied consistently across all conditions and repeats.

B.1.7 Random seeds and reproducibility

Each run used a unique per-run seed generated without fixed seeding to avoid correlations across parallel workers. The seed combined a high-resolution time component with the process ID and an additional random draw, then reduced modulo $2^{31} - 1$. Each run stored its seed (argument `seed`) within the run history and metadata for reproducibility.

B.1.8 Run outputs and file naming

Each optimisation produced two outputs:

- `H`: the full history structure containing iteration-wise geometry (`iter(k).M`), objective (`iter(k).J`), statistics (`iter(k).stats`), and grid points (`iter(k).gridPts`), as well as start/final metadata in `H.meta`.
- `runInfo`: a lightweight struct with identifiers (configuration tag, repeat index, start mode, CSV stem if applicable).

Files were saved as individual `.mat` files to `runs/` with filenames encoding condition, final microphone count, repeat index, and timestamp.

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B.2 Analysis pipeline and figure generation

All batch runs were analysed using `iwah_analyse_batch_runs.m`, which loads all `.mat` files in `runs/` and extracts common summary products. Unless stated otherwise, I report **position error** (`METRIC='pos'`) in cm. Below I describe each analysis component in the same order as the script, and I indicate the corresponding figures produced by the pipeline.

B.2.1 Run ingestion, grouping, and "before" vs "after" selection

For each run file, I identified:

- microphone count N_{mic} as the number of rows in `H.meta.final.M` (fallback: `H.meta.start.M`);
- start type (`csv` vs `random`) using `H.meta.start.csvPath` (or fallback string parsing of `H.meta.start.source`);
- condition key as the CSV filename stem for CSV starts and as `randomStart` for random starts; and
- a short label for each condition using a fixed mapping (`COND_LABELS`).

I defined **"before"** as the first recorded iteration (`iBase=1`). I defined **"after"** using one of two modes:

1. `AFTER_MODE='final'`: the last recorded iteration (`iFinal=numel(H.iter)`), or
2. `AFTER_MODE='best'`: the iteration achieving the maximum pass rate, with ties broken by (i) smaller p_{95} error and then (ii) larger objective value J.

Masking of unused test points. Errors were optionally computed only over valid/used test points (`IGNORE_UNUSED=true`) by applying the boolean vector `stats.usedMask`. This step excludes grid points that the localisation/evaluation routine marks as unused (e.g. invalid due to constraints or internal screening) and prevents unused points from biasing histograms and summary statistics.

B.2.2 Figure: pass-rate trajectories across iterations

This figure summarises optimisation dynamics using pass-rate trajectories. For each $N_{\mathrm{mic}} \in \{4, 6, 8, 10, 12\}$ I grouped runs by condition (CSV geometry or random start) and computed a robust central trajectory:

1. I extracted iteration-wise pass rates `iter(k).stats.passRate`.
2. Only **integer** iteration indices were retained to exclude shake steps (e.g. $it = 0.5$) and to ensure consistent alignment.
3. Each run trajectory was aligned onto a common integer iteration axis by forward-filling missing iterations (holding the last observed value).
4. I summarised across repeats using the median trajectory and a shaded uncertainty band, using either the interquartile range (IQR: 25–75%) or a 10–90% band (parameter `TRAJ.band`).

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B.2.3 Figure: "average-run" error distributions before vs after

This figure visualises how optimisation redistributes localisation error. For each N_{mic} and start type (CSV vs random), I computed filled histograms for pre- and post-optimisation errors:

1. For each run, I extracted the error vector `stats.err_cm` (or `stats.ang_deg`) at the "before" and "after" iterations.
2. If enabled, I filtered these vectors using `stats.usedMask`.
3. To avoid histogram domination by extreme failures, I optionally capped errors at a multiple of the pass threshold (`HIST.capMode`: `2xThreshold` or `4xThreshold`).
4. Within each panel, histogram bin edges were determined from the pooled (capped) values to ensure a consistent binning scheme across "before" and "after".
5. The displayed histogram is an *average-run* distribution: for each run I computed a probability histogram (normalised to sum to 1), then averaged these probabilities across repeats.

A special case was applied for $N_{\mathrm{mic}} = 4$ CSV starts: Pyramid and Tetrahedron conditions were plotted separately (distinct colours) to allow direct comparison of the two 4-microphone canonical geometries within the same panel.

B.2.4 Figure: accuracy vs distance from array centre

This figure quantifies spatial structure in localisation error by summarising error as a function of radial distance from the array centre. For each run I computed the radial coordinate $r = \|\mathbf{x}\|_2$ for each grid point `gridPts`. I then:

1. defined $n_r = 10$ radial bins spanning the range of r values in the dataset;
2. within each bin, computed robust statistics for "before" and "after" errors: median and a quantile band (default 10–90%, parameters `RAD.qLo`, `RAD.qHi`);
3. optionally applied an error cap (default `RAD.capForScatter='2xThreshold'`) to reduce the influence of catastrophic failures on the quantile summaries; and
4. aggregated across repeats by taking the median of run-wise medians (and of run-wise quantiles) at each radial bin centre.

B.2.5 Evaluation of before–after differences and robust statistics

Across all analyses, pre- and post-optimisation differences were evaluated using robust, distribution-aware summary statistics computed on masked error vectors. For each run, error values were first filtered using the boolean vector `stats.usedMask`, thereby excluding grid points that were deemed invalid or unused by the localisation routine (e.g. due to geometric infeasibility or internal screening). All reported statistics are therefore conditioned on the same set of valid test points before and after optimisation.

To characterise localisation performance beyond the mean, I focused on upper-tail behaviour using the 95th percentile of the error distribution (p_{95}). This choice reflects worst-case but non-pathological performance and avoids sensitivity to extreme outliers, which are handled separately (see below). For each configuration group, I computed:

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- pass rate (fraction of points with $e \leq \theta$),
- mean error,
- median error,
- interquartile range (IQR; 75th–25th percentile), and
- p_{95} error.

These quantities were computed for both "before" and "after" states, and differences were defined as $\Delta x = x_{\text{after}} - x_{\text{before}}$.

B.2.6 Figure: representative geometry and error-field visualisation

This figure provides a qualitative illustration of how optimisation alters both microphone geometry and the resulting spatial error field. For a selected canonical configuration (specified by `SELECT_GEOM`), I identified a representative run as the one achieving the highest post-optimisation pass rate, with ties broken by the smallest p_{95} error.

For this run, I visualised:

- the microphone coordinates at the initial state and after optimisation,
- the three-dimensional test grid coloured by localisation error,
- the outer evaluation sphere (R) and maximum geometry extent ($L_{\max}$), and
- the array centre (origin).

Error values were capped at 2× the pass threshold to stabilise colour scaling and highlight spatial structure in the bulk of the distribution. Only grid points marked as valid by `usedMask` were included. This figure serves as an intuitive complement to the statistical summaries presented elsewhere.

B.2.7 Pooled summary statistics table

In addition to the figures, I computed pooled summary statistics across runs for each configuration group. For CSV starts, statistics were stratified by canonical geometry and microphone count; for random starts, statistics were stratified by microphone count only. Error vectors were pooled across all repeats within each group after applying `usedMask`.

For each group, I report pass rate, mean, median, IQR, and p_{95} error for both pre- and post-optimisation states. These pooled values form the basis for quantitative comparisons across configurations and array sizes.

B.2.8 Delta mosaic of optimisation effects

To compactly summarise the direction and magnitude of optimisation effects across all configurations, I constructed a *delta mosaic* heatmap. Each row corresponds to a configuration (canonical geometries and random starts), and each column corresponds to a summary metric (pass rate, mean, median, IQR, p_{95}, and optional tail fraction).

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For interpretability, deltas were displayed such that positive values indicate improvement. Specifically, pass-rate deltas were used directly (Δ in percentage points), whereas error-based deltas were sign-inverted so that reductions in error appear as positive values.

Colour scaling was made robust by setting symmetric limits based on the 95th percentile of $|\Delta|$ across all cells. This prevents a small number of extreme changes from dominating the visual encoding while preserving relative differences among configurations.

B.2.9 Tail behaviour and catastrophic errors

To explicitly track severe localisation failures, I quantified the fraction of grid points with errors exceeding a multiple of the pass threshold. Errors larger than $4\times\theta$ were classified as catastrophic. For each configuration group, I computed the percentage of such tail points before and after optimisation and included the corresponding change (Δ tail fraction) as an optional column in the delta mosaic.

This separation allows improvements in the bulk of the distribution (median, IQR) to be interpreted independently from changes in extreme failure rates.

B.2.10 Optimisation-induced geometry drift

To quantify how much the optimiser altered microphone positions, I analysed geometry drift using `iwah_geometry_drift_plots_etc.m`. For each run, I extracted start and end microphone coordinates and computed per-microphone displacement vectors

$$\Delta\mathbf{m}_i = \mathbf{m}_{i,\text{after}} - \mathbf{m}_{i,\text{before}}. \tag{12}$$

Drifts were pooled across microphones and repeats within each configuration and visualised as boxplots for the x, y, and z components separately. A horizontal reference line at zero indicates no net displacement.

This analysis allows assessment of whether performance gains arise from large geometric rearrangements or from comparatively subtle adjustments.

B.2.11 Runtime diagnostics for pass-rate discrepancies

To explain why different start geometries can exhibit different *achievable* maximum pass rates (and not merely different convergence speeds), I computed a set of per-run runtime diagnostics from the full optimiser history (`H.iter`) and summarised these by configuration. Diagnostics were extracted from integer iteration indices only (i.e. excluding non-integer "shake" steps) to match the trajectory alignment used in Figure 1.

For each run, I derived the pass-rate trajectory $\{p(it)\}$ from `iter(k).stats.passRate`. If the stored values were fractions (0–1), they were converted to percentages for reporting. I then computed:

- **Achievable max pass rate:** $p_{\max} = \max_{it} p(it)$.
- **Final pass rate:** $p_{\text{final}} = p(it_{\text{end}})$, and **net change** $\Delta p = p_{\text{final}} - p(it{=}1)$.
- **Convergence iteration:** the first iteration at which the run reached a fixed fraction of

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its own maximum,

$$it_{\text{conv}} = \min\{it : \; p(it) \geq \eta \, p_{\max}\}, \qquad \eta = 0.95, \tag{13}$$

thereby separating differences in *attainable* performance from differences in *speed* of improvement.

- **Step-size proxy:** a per-iteration geometry update magnitude computed from successive microphone-coordinate matrices,

$$s(it) = \sqrt{\frac{1}{3N} \sum_{i=1}^{N} \|\mathbf{m}_i(it) - \mathbf{m}_i(it{-}1)\|_2^2}, \tag{14}$$

and summarised per run by the mean step size $\bar{s}$ and the final step size s_{last}. This quantifies how strongly the optimiser had to move the array to achieve its final state.

- **Net drift magnitude:** an overall displacement scale computed from the start and "after" geometries,

$$\text{drift}_{\text{RMS}} = \sqrt{\frac{1}{3N} \sum_{i=1}^{N} \|\mathbf{m}_{i,\text{after}} - \mathbf{m}_{i,\text{before}}\|_2^2}. \tag{15}$$

Diagnostics were aggregated by configuration ID using the median across runs (with IQR where appropriate). I additionally visualised $p_{\max}$, p_{final}, and it_{conv} as grouped boxplots and performed pooled correlation checks between $p_{\max}$ and the step/drift proxies (to test whether higher attainable pass rates were systematically associated with larger geometry movements). These diagnostics provide an interpretable separation between (i) geometry-dependent limits on achievable localisation performance under the fixed test grid and constraints, and (ii) optimisation dynamics such as convergence speed and stability.